

Nejdříve dokážeme $\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$

Podle definice normy je $\|\vec{u} + \vec{v}\|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle$

$$\|\vec{u} - \vec{v}\|^2 = \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle$$

To díky rovnostem ze skript (Definice 8.14, Poznámka 8.15) lze rozepsat:

$$\begin{aligned}\|\vec{u} + \vec{v}\|^2 &= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)\end{aligned}$$

V poslední rovnosti jsme využili toho, že pro $x \in \mathbb{C}$ platí $x + \bar{x} = 2\operatorname{Re}(x)$.

(Lze ukázat jako $x = a + bi$
 $\bar{x} = a - bi \rightarrow x + \bar{x} = a + bi + a - bi = 2a$)

Podobně rozepíšeme $\|\vec{u} - \vec{v}\|^2$:

$$\begin{aligned}\|\vec{u} + (-\vec{v})\|^2 &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -\vec{v} \rangle + \langle -\vec{v}, \vec{u} \rangle + \langle -\vec{v}, -\vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 - \langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)\end{aligned}$$

Dohromady:

$$\begin{aligned}\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 &= 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) - (-2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)) = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \\ \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) &= \frac{1}{4}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)\end{aligned}$$

Podobně odvodíme $\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$.

Opět z definice normy a rovností z definice 8.14, pozorování 8.15 dostáváme

$$\begin{aligned}\|\vec{u} + i\vec{v}\|^2 &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle = \\ &= \langle \vec{u}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle - i\langle \vec{v}, \vec{u} \rangle - i^2 \langle \vec{v}, \vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + i\langle \vec{u}, \vec{v} \rangle - i\overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)\end{aligned}$$

V poslední rovnosti jsme využili, že $ix - i\bar{x} = 2\operatorname{Im}(x)$

(Lze ukázat jako $x = a + bi$, $\bar{x} = a - bi$

$$ix - i\bar{x} = ia - b - ia - b = -2b)$$

Podobně:

$$\begin{aligned}\|\vec{u} - i\vec{v}\|^2 &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -i\vec{v} \rangle + \langle -i\vec{v}, \vec{u} \rangle + \langle -i\vec{v}, -i\vec{v} \rangle = \\ &= \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle - i^2 \langle \vec{v}, \vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 - i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)\end{aligned}$$

Dohromady:

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 = 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) - (-2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle))$$

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$